USER INTERFACE DESIGN EXP: 9

Design input forms that validate data (e.g., email, phone number) and display error messages using HTML/CSS, JavaScript (with Validator.js)

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AIM:

To design and develop an input form that validates user data such as email and phone number, and displays appropriate error and success messages using HTML, CSS, and JavaScript.

DESCRIPTION:

This project involves creating a user-friendly form that accepts an email and phone number. The form validates:

- Email format (must be valid)
- Phone number (must be exactly 10 digits)

If invalid data is entered, error messages are displayed in red.
If both inputs are valid, a success message is displayed.

PROCEDURE:

Step 1: Create HTML Structure

- Create a form with:
  - Email input field
  - Phone number input field
  - Submit button
- Add separate areas to display error messages.

Step 2: Apply CSS Styling

- Style the form with:
  - Center alignment
  - Input box spacing
  - Rounded borders
  - Highlighted focus effect
- Use:
  - Red color for error messages
  - Green box/message for success

Step 3: Add JavaScript Validation

- Prevent form submission using `event.preventDefault()`
- Validate:
  - Email using regex or Validator.js
  - Phone number length (must be 10 digits)
- Display:
  - Error messages if invalid
  - Success message if both are valid

ALGORITHM:

1. Start
2. Get email and phone input values
3. Check if email is valid
   - If not, display email error
4. Check if phone number is 10 digits
   - If not, display phone error
5. If both are valid
   - Display success message
6. Stop

PROGRAM:

HTML CODE:

```html
index.html > ...
<!DOCTYPE html>
older to start working (⌘O)
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Form Validation - No Browser Tooltips</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <main class="container">
        <form id="validationForm" novalidate>
            <div class="form-group">
                <label for="email">Email:</label>
                <input type="text" id="email" placeholder="Enter your email">
                <span id="emailError" class="error"></span>
            </div>
            <div class="form-group">
                <label for="phone">Phone Number:</label>
                <input type="text" id="phone" placeholder="Enter your phone number">
                <span id="phoneError" class="error"></span>
            </div>
            <button type="submit">Submit</button>
            <p id="successMessage"></p>
        </form>
    </main>
    <script src="https://cdnjs.cloudflare.com/ajax/libs/validator/13.6.0/validator.min.js"></script>
    <script src="script.js"></script>
</body>
</html>
```

CSS CODE:

```css
* {
    box-sizing: border-box;
}

html, body {
    margin: 0;
    padding: 20px;
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    font-family: 'Inter', -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif;
}

.container {
    max-width: 420px;
    width: 90%;
    background: white;
    padding: 2.5rem;
    border-radius: 16px;
    box-shadow: 0 20px 40px rgba(0,0,0,0.1);
}

.form-group {
    margin-bottom: 1.75rem;
}

label {
    display: block;
    margin-bottom: 0.5rem;
    font-weight: 600;
    color: #333;
    font-size: 0.95rem;
}
```

Open a file or folder to start working (⌘O)

```css

input[type="text"] {
    width: 100%;
    padding: 12px 16px;
    border: 2px solid #ddd;
    border-radius: 8px;
    font-size: 1rem;
    transition: border-color 0.3s, box-shadow 0.3s;
}

input:focus {
    outline: none;
    border-color: #4299e1;
    box-shadow: 0 0 0 3px rgba(66, 153, 225, 0.1);
}

/* Standard RED error text below inputs */
.error {
    color: red;
    font-size: 0.85em;
    margin-top: 5px;
    display: block;
    min-height: 20px; /* Reserve space to prevent layout jump */
}

/* Green success message */
#successMessage {
    margin-top: 1rem;
    padding: 12px 16px;
    background-color: #48bb78;
    color: white;
    border-radius: 8px;
    font-size: 0.9rem;
    font-weight: 500;
    text-align: center;
```

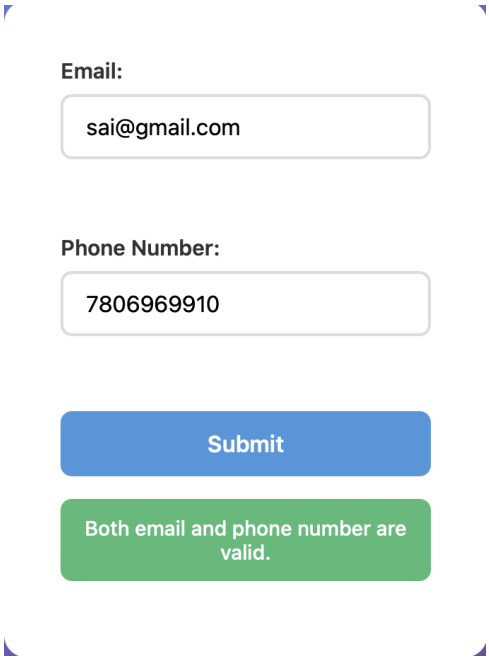
```
    text-align: center;
    display: none;
}

button[type="submit"] {
    width: 100%;
    padding: 14px;
    background: #4299e1;
    color: white;
    border: none;
    border-radius: 8px;
    font-size: 1rem;
    font-weight: 600;
    cursor: pointer;
    transition: background 0.3s, transform 0.2s;
}

button:hover {
    background: #3182ce;
    transform: translateY(-1px);
}

button:disabled {
    background: #a0aec0;
    cursor: not-allowed;
    transform: none;
}

/* Responsive */
@media (max-width: 480px) {
    .container {
        margin: 1rem;
        padding: 2rem;
    }
}
```

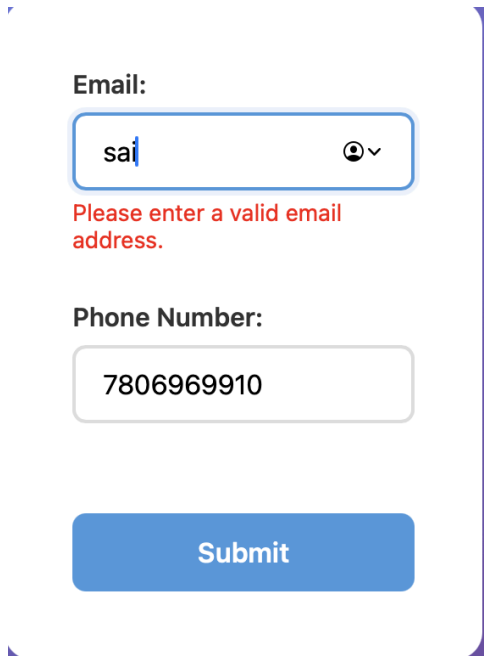
JAVASCRIPT CODE:

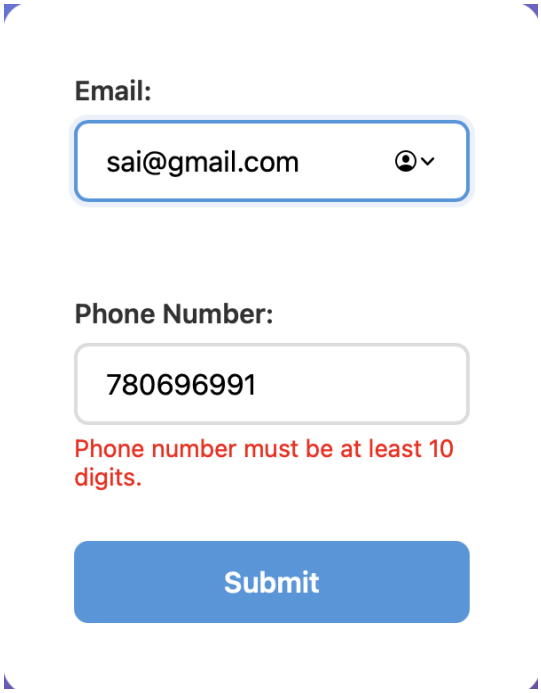
```javascript
document.addEventListener('DOMContentLoaded', function() {
    const form = document.getElementById('validationForm');
    const emailInput = document.getElementById('email');
    const phoneInput = document.getElementById('phone');
    const emailError = document.getElementById('emailError');
    const phoneError = document.getElementById('phoneError');
    const successMessage = document.getElementById('successMessage');
    const submitBtn = form.querySelector('button');

    /**
     * Clear all messages on submit
     */
    function clearMessages() {
        emailError.textContent = '';
        phoneError.textContent = '';
        successMessage.style.display = 'none';
    }

    /**
     * Show red error text below specific input using textContent
     */
    function showFieldError(errorElement, message) {
        errorElement.textContent = message;
    }

    /**
     * Show green success message
     */
    function showSuccess() {
        successMessage.textContent = 'Both email and phone number are valid.';
        successMessage.style.display = 'block';
    }

    /**
     * Phone validation: format + 10+ digits
     */
    function validatePhone(phone) {
        if (!validator.isMobilePhone(phone)) {
            return 'Please enter a valid phone number.';
        }
        const digitsOnly = phone.replace(/\D/g, '');
        if (digitsOnly.length < 10) {
            return 'Phone number must be at least 10 digits.';
        }
        return true;
    }

    // Form submit - ALL JS validation, no browser interference
    form.addEventListener('submit', function(e) {
        e.preventDefault(); // No page reload

        clearMessages();
        let isValid = true;

        // Email validation
        const email = emailInput.value.trim();
        if (!validator.isEmail(email)) {
            showFieldError(emailError, 'Please enter a valid email address.');
            isValid = false;
        }

        // Phone validation
        const phone = phoneInput.value.trim();
        const phoneResult = validatePhone(phone);
        if (phoneResult !== true) {
            showFieldError(phoneError, phoneResult);
            isValid = false;
        }

        // Success ONLY when both valid
        if (isValid) {
            showSuccess();
        }
    });

    // Clear errors on input focus (UX)
    emailInput.addEventListener('focus', () => emailError.textContent = '');
    phoneInput.addEventListener('focus', () => phoneError.textContent = '');
});
```

OUTPUT:

- Displays error message when:
  - Email is invalid
  - Phone number is less than 10 digits
- Displays success message:
  - “Both email and phone number are valid.” when inputs are correct







RESULT:

Thus, the input form was successfully designed to validate email and phone number and display appropriate error and success messages using HTML, CSS, and